Lilian William Hollard

Ph.D. Candidate in Machine Learning

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Professional Summary

Ph.D. researcher in Machine Learning with over 3 years of experience in optimizing deep neural network architectures, specializing in CNNs, Transformers, real-time object detection, and image classification. Proven track record in reducing model complexity while maintaining high accuracy for industrial applications.

Education

- **University of Reims Champagne Ardenne** — *Reims, France*
  Ph.D. in Machine Learning, Lightweight Vision Models for Embedded Systems — *Feb. 2023 – Expected Nov. 2025*
- **University of Reims Champagne Ardenne** — *Reims, France*
  Master of Science in Computer Science — *Sep. 2020 – June 2022*

Professional Experience

- **LICIIS Laboratory** — *Reims, France*
  Machine Learning Researcher — *Nov. 2022 – Present*
  - **Neural Network Architecture Design**: Develop, implement, and optimize deep neural networks for Computer Vision applications, achieving significant cost reduction while improving accuracy using PyTorch framework.
  - **CNN Performance Optimization**: Reduced YOLO model inference time by 26% for industrial applications with zero accuracy loss compared to state-of-the-art object detectors.
  - **Transformer Architecture Innovation**: Engineered one of the lightest Vision Transformers for classification (0.9M parameters, 70% ImageNet accuracy), optimizing costly architectures for resource-constrained environments.
  - **Infrastructure Development**: Built comprehensive analytics environment using Docker, AWS, and Romeo Cloud; authored core Computer Vision libraries adopted across research teams.
  - **Research Publications**: Authored multiple first-author papers in English and French for peer-reviewed international conferences.
  - **Project Leadership**: Managed complete ML lifecycle from raw data processing to deployment-ready model implementation.
  - **AI Production Systems**: Designed and implemented AI pipelines for vineyard yield estimation, significantly improving harvest efficiency through precision agriculture.
  - **Grant Acquisition & Strategy**: Contributed to proposal writing, funding acquisition, and EU grant documentation for multi-million euro research projects.
  - **Applied Computer Vision**: Developed real-time object detection systems for vineyard monitoring, enabling embedded deployment in agricultural robotics.
  - **Industry Collaboration**: Facilitated technical alignment between academic researchers and industry stakeholders through regular workshops and demonstrations.
- **University of Reims Champagne Ardenne (URCA)** — *Reims, France*
  Associate Professor — *Sep. 2024 – April 2025*
  - **Advanced Teaching**: Delivered graduate-level Deep Learning courses, designed practical neural network implementations in PyTorch, and organized hands-on model training workshops.
  - **Curriculum Development**: Created comprehensive teaching materials for Algorithms and C programming courses for second-year undergraduates.

Publications [Google Scholar]

- **LeYOLO: New Scalable and Efficient CNN for Object Detection**: Accepted at CRV 2025 – arXiv:2406.14239
- **Exploring Superposition in Low-Param Vision Models**: Canadian AI Conference (CAIAC), 2025 – Publication Link
- **Smart-Viticulture and Deep Learning: Challenges and Recent Developments on Yield Prediction**: Smart Life and Smart Life Engineering, Springer Nature 2025 - Springer Link
- **Improving Edge-AI Image Classification Through the Use of Better Building Blocks**: CLOSER 2024 - Conference Paper
- **Machine Learning Forecast of Soybean Yields in South Brazil**: Workshop at IE2022 - IOS Press
- **Applying Knowledge Distillation on Pre-Trained Model for Early Grapevine Detection**: Workshop at IE2023 - DOI Link

Technical Expertise

- **Programming Languages**: Python, C, C++ **Technologies & Frameworks**: AWS, Docker, Singularity, PyTorch, Scikit-Learn
- **Core Competencies**: Deep Learning, Computer Vision, Neural Architecture Search, Model Optimization, Real-time Inference, Embedded AI, Agricultural Technology, Research Leadership

References

- **Luiz-Angelo Steffenel**: Professor - luiz-angelo.steffenel@univ-reims.fr
- **Lucas Mohimont**: Postdoctoral Fellow - lucas.mohimont@univ-reims.fr
- **Nathalie Gaveau**: Professor - nathalie.gaveau@univ-reims.fr